

The contribution of the MSA to $\Delta\Delta G$ prediction

Experiment 04 · ddG_with_Boltz · raw- Δz features · gradient-boosted trees · MSA vs single-sequence.

Running the structure model in **single-sequence mode** (no multiple-sequence alignment) costs a strikingly **uniform $\sim 0.08\text{-}0.10$ pooled Pearson r** on every holdout (mean $\Delta r \approx -0.086$). Two conclusions follow: most of the $\Delta\Delta G$ signal is **structural** — single-sequence Boltz still reaches $r \approx 0.70$ — and the MSA adds a real, non-trivial boost on top, largest for de-novo transfer. The predictor degrades gracefully, not catastrophically, without the MSA.

1. Motivation

The structure model normally sees a multiple-sequence alignment (MSA) per protein — an evolutionary signal that is itself informative about stability. Removing it isolates what the model's structural prior alone contributes, and tells us whether the pipeline depends on a rate-limited, sometimes-unavailable alignment server.

2. Methods

Identical corpus, features, and model to the generalization study — the **only** difference is the MSA. Single-sequence mode makes the pipeline emit an empty alignment in every query; everything downstream (raw Δz features, gradient-boosted trees, the full holdout suite) is unchanged. Both runs use the 256 z-only raw- Δz features, compared holdout-by-holdout.

3. Results

Holdout	MSA	no-MSA	Δr
random	0.783	0.699	-0.084
protein	0.774	0.687	-0.087
denovo	0.705	0.606	-0.099
substitution	0.772	0.691	-0.081
source_residue	0.754	0.670	-0.085
target_residue	0.743	0.664	-0.079
chemistry	0.734	0.647	-0.087

Table 1. Pooled Pearson r , MSA vs single-sequence, per holdout.

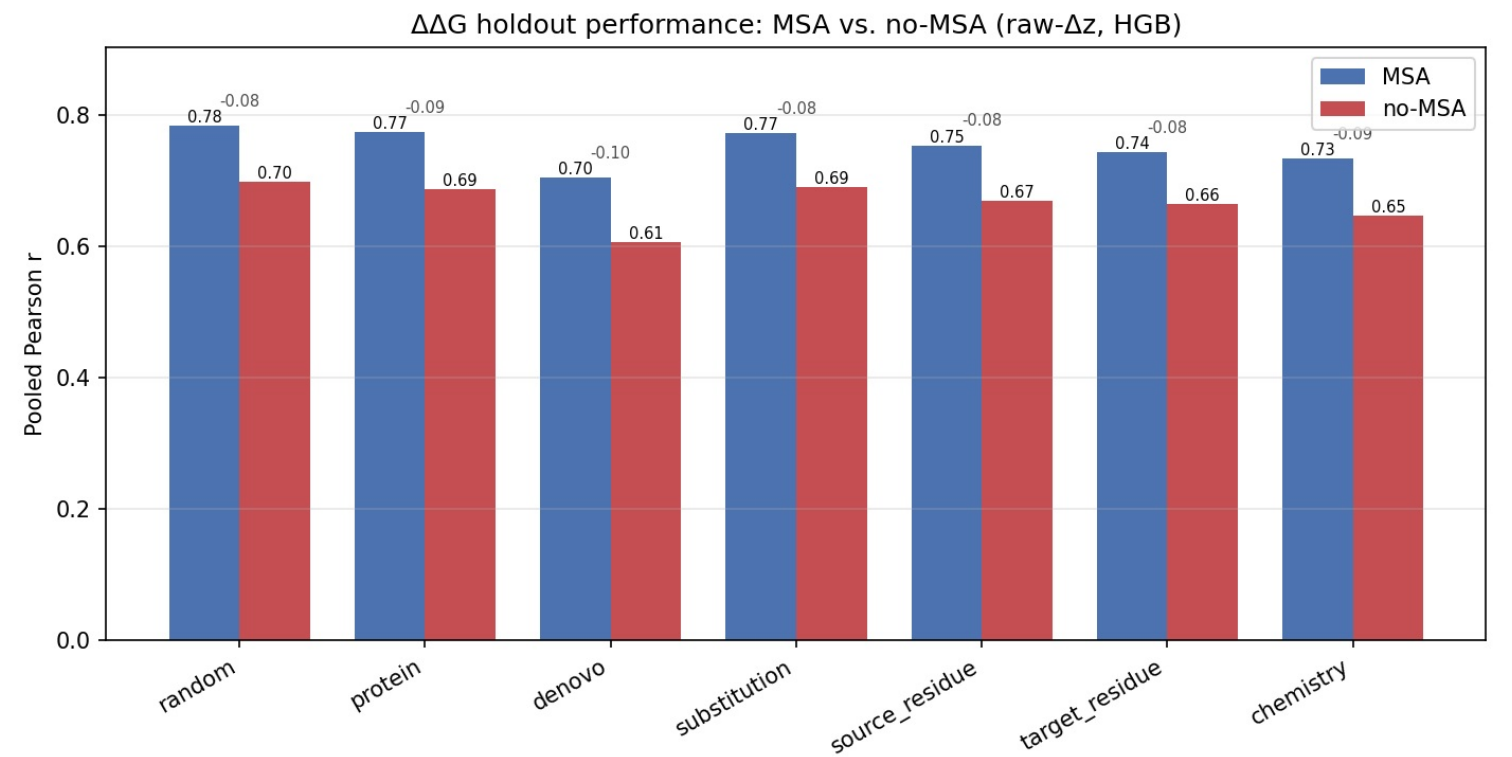


Figure 1. Pooled Pearson r , MSA vs single-sequence Boltz per holdout, with Δr labels.

4. Discussion

The penalty for removing the MSA is remarkably consistent — about -0.086 r and ~ 0.09 kcal/mol RMSE across every holdout — rather than concentrated in one split. Its largest effect is on **de-novo transfer** (-0.099): predicting designed proteins leans most on the evolutionary input. Even so, single-sequence Boltz still reaches $r \approx 0.70$ (random) and 0.69 (unseen proteins), so the structural prior carries the bulk of the signal. In practice: keep the MSA when it is available for the extra boost, but the method remains useful without it.

